

Project Initialization and Planning Phase

Date	29 June 2026
Team ID	xxxxxx
Project Title	A Comprehensive Measure of Well-Being
Maximum Marks	5 Marks

Project Proposal (Proposed Solution) report

The proposal report aims to transform loan approval using machine learning, boosting efficiency and accuracy. It tackles system inefficiencies, promising better operations, reduced risks, and happier customers. Key features include a machine learning-based credit model and real-time decision-making.

Project Overview	
Objective	The primary objective is to build an intelligent HDI prediction system using Linear Regression that allows users to instantly predict a country's Human Development Index score by entering four simple indicators — Life Expectancy, Expected Years of Schooling, Mean Years of Schooling, and GNI per Capita .
Scope	The project covers data collection from Kaggle, data preprocessing, model training and evaluation, model serialization using Pickle, and deployment as a Flask web application with three HTML pages — home, prediction form, and result page .
Problem Statement	

Description	No simple, interactive, ML-powered tool currently exists for predicting HDI scores from basic country indicators. Existing resources such as UNDP PDF reports and Excel spreadsheets require significant time and technical expertise, making HDI analysis inaccessible to most users
Impact	The absence of such a tool forces policy makers, researchers, and students to spend hours manually analyzing raw data. This delays decision-making and reduces the efficiency of development planning and research
Proposed Solution	
Approach	Using Linear Regression trained on UNDP 2021 HDI dataset from Kaggle covering 191 countries, the model learns the relationship between key human development indicators and HDI scores. The trained model is saved using Pickle and integrated into a Flask web application for real-time predictions
Key Features	Real-time HDI score prediction from 4 input indicators. - Live HDI score estimator that updates as user types. - Automatic classification into Low / Medium / High / Very High HDI. - Dark-themed interactive web UI with progress tracker. - Lightweight Flask app accessible on any browser

Resource Requirements:

Resource Type	Description	Specification/Allocation
Hardware		
Computing Resources	CPU/GPU specifications, number of cores	Intel Core i3 or above
Memory	RAM specifications	Minimum 4 GB RAM
Storage	Disk space for data, models, and logs	Minimum 10 GB free space
Software		
Frameworks	Python web framework	Flask 2.x
Libraries	ML and data libraries	scikit-learn, pandas, numpy, matplotlib, seaborn
Development Environment	IDE and notebook	VS Code, Jupyter Notebook
Data		
Data	Source, size, format	Kaggle UNDP HDI Dataset, 191 rows × 82 columns, CSV format